Capacitive-Type Flexible Linear Encoder With Untethered Slider Using Electrostatic Induction

Fumitaka Kimura, *Student Member, IEEE*, Masahiko Gondo, Norio Yamashita, *Student Member, IEEE*, Akio Yamamoto, *Member, IEEE*, and Toshiro Higuchi, *Member, IEEE*

***Abstract*—A novel capacitive linear encoder characterized by its untethered slider is presented. The main components are made of flexible printed circuit films measuring 0.2 mm in thickness that allows the sensor to be set up in thin interspaces or on curved surfaces. The sensor consists of a long receiver film and a short transmitter film, respectively containing four-phase and two-phase electrodes; the transmitter is used as a slider and the receiver as a stator. To realize an untethered slider, the sensor employs a unique approach; electric power is supplied to the transmitter film by electrostatic induction, which removes electric wires from the slider. This untethered slider can facilitate sensitive applications where a mechanical disturbance caused by an electric wire can be a problem. The principle was verified using a prototype encoder, which showed an error of ± 4 μm.**

***Index Terms*—Capacitive encoder, incremental encoder, linear displacement sensor, linear encoder, wireless slider.**

I. Introduction

LINEAR encoders are widely used in various positioning applications, such as numerical controlled (NC) machine tools and factory automation, since they are essential for precision positioning systems [1]–[7]. Recently, the needs for linear encoders have been growing with increased needs for automation. Under such circumstances, the demand for a new encoder, which is more robust and low-cost, has also been growing. Typical linear encoders can be classified into three types: the optical type, the magnetic type and the capacitive type. Optical encoders are one of the most common instruments in precision positioning devices since they have a large measuring displacement with a very fine resolution such as several nanometers, if special glass or metal scales are used. However, they require higher cost for glass or metal scale production and a larger space for an installation and an operation. In addition, special care should be taken when they are used in dirty environments since their performance can be easily affected by dust. On the other hand, magnetic encoders have robust structures and high reliability for diverse environments. However, their resolutions and accuracies are typically lower than those of optical encoders. Finally, capacitive encoders are mainly used in small components, e.g., a digital vernier caliper due to its simple structure, high sensitivity and robustness to changes in temperature and a magnetic field [8]–[11].

Manuscript received September 16, 2009; accepted October 29, 2009. Current version published March 26, 2010. This work was supported in part by the Creation and Support Program for Start-Ups from Universities from the Japan Science and Technology Agency of Japan. This is an expanded paper from the IEEE SENSORS 2008 Conference. The associate editor coordinating the review of this paper and approving it for publication was Dr. Patrick Ruther.

The authors are with the Department of Precision Engineering, the University of Tokyo, Tokyo 113-8656, Japan (e-mail: k-bunbun@aml.t.u-tokyo.ac.jp).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/JSEN.2009.2037015

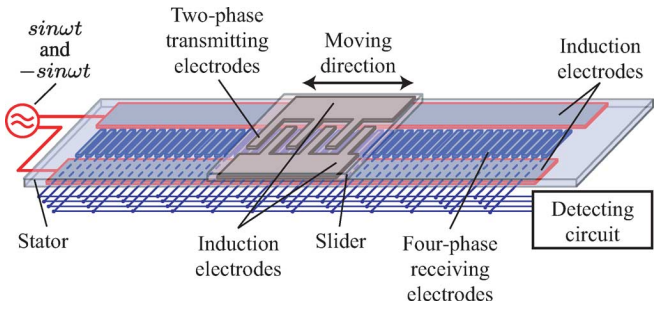


Fig. 1. Schematic diagram of electrostatic capacitive encoder.

Reviewing the existing encoders, it can be found that, regardless of their working principles, conventional encoders typically have cabling in their sliding parts. In general, a scale has a length equal to or longer than the stroke of the linear motion and is naturally attached to the stationary part of the system. Then, the sliding part is equipped with a sensor head, which has cabling for signal transmission. In some applications, such cabling can be a fatal problem; for example, force sensitive systems may suffer from the mechanical disturbance caused by the cabling, or high-speed systems can have a problem regarding breakings of wires. In addition, the cabling to the sliding parts can make the system structure complicated.

In this paper, we developed a novel capacitive encoder that is characterized by its untethered slider [12], [13]. Typically, capacitive encoders utilize a pair of electrodes, one of which is used as a transmitter and the other is used as a receiver [8], [14]. Normally both the stator and the slider require cabling, and therefore a sliding part needs to be tethered. Although there are some encoders that realize untethered slider [15], such untethered encoders have not been extensively studied. This paper proposes a new principle to realize an untethered slider by utilizing electrostatic induction. In the next section, the paper describes the proposed structure and its analytical model. In Section III, the optimized design of the electrodes will be discussed from the viewpoint of the output error. Then, a prototype encoder will be introduced in Section IV, which will be tested to verify the proposed principle in Section V.

II. Basic Principle

A. Structure

Fig. 1 shows the schematic diagram of the proposed capacitive encoder. It mainly consists of two parts: a stator working

1530-437X/$26.00 © 2010 IEEE
Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

as a receiver and a slider as a transmitter. The slider has a two-phase parallel electrode on its center. This two-phase electrode will be energized to form a potential field using two-mirrored sine waves that have 180-degree phase shift. The potential field is detected by the four-phase electrode formed on the stator. The two parts are aligned so that slider's two-phase electrode slides over the stator's four-phase electrode. The pitches of electrodes are p in the slider and $2p$ in the stator.

In addition to those electrodes, both the stator and the slider have additional electrodes along their long sides, which are used as induction electrodes. They are designed so that the induction electrodes on the slider always overlap those on the stator during operation. The induction electrodes on the stator are energized by the two-mirrored sine waves during operation. Since the induction electrodes on the slider and on the stator form capacitors, the application of ac voltages to the stator induction electrodes can induce ac electric potentials on the slider induction electrodes. Since the slider induction electrodes are connected to the two-phase electrode, electric energy can be transferred to the two-phase electrode without any cabling.

B. Analytical Model

Fig. 2 shows an analytical equivalent model for this encoder [16]. Since this encoder shares the same electrode structure as the electrostatic actuator described in [16], the same equivalent model is used for the analysis. In this model, the encoder structure is represented by eight terminals; the terminals 1 through 4 correspond to the four-phase electrodes on the stator, the terminals 5 and 6 correspond to the two-phase electrodes, which are connected to the slider induction electrodes. Then, the terminals 7 and 8 correspond to the stator induction electrodes on the stator.

Among the electric terminals, capacitances are formed between any pairs of terminals. Such capacitances can be mathematically represented using a capacitance coefficient matrix [17]. Considering the geometrical symmetricity among terminals, the matrix for this encoder can be expressed as [16]

$$\mathbf{C} = \begin{bmatrix} C_{\mathrm{st}} & -C_{\mathrm{ta}} & -C_{\mathrm{tb}} & -C_{\mathrm{ta}} \\ -C_{\mathrm{ta}} & C_{\mathrm{st}} & -C_{\mathrm{ta}} & -C_{\mathrm{tb}} \\ -C_{\mathrm{tb}} & -C_{\mathrm{ta}} & C_{\mathrm{st}} & -C_{\mathrm{ta}} \\ -C_{\mathrm{ta}} & -C_{\mathrm{tb}} & -C_{\mathrm{ta}} & -C_{\mathrm{st}} \\ C_{\mathrm{m}}(\theta_x) & C_{\mathrm{m}}\left(\theta_x-\frac{\pi}{2}\right) & C_{\mathrm{m}}(\theta_x-\pi) & C_{\mathrm{m}}\left(\theta_x+\frac{\pi}{2}\right) \\ C_{\mathrm{m}}(\theta_x-\pi) & C_{\mathrm{m}}\left(\theta_x+\frac{\pi}{2}\right) & C_{\mathrm{m}}(\theta_x) & C_{\mathrm{m}}\left(\theta_x-\frac{\pi}{2}\right) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} C_{\mathrm{m}}(\theta_x) & C_{\mathrm{m}}(\theta_x-\pi) & 0 & 0 \\ C_{\mathrm{m}}\left(\theta_x-\frac{\pi}{2}\right) & C_{\mathrm{m}}\left(\theta_x+\frac{\pi}{2}\right) & 0 & 0 \\ C_{\mathrm{m}}(\theta_x-\pi) & C_{\mathrm{m}}(\theta_x) & 0 & 0 \\ C_{\mathrm{m}}\left(\theta_x+\frac{\pi}{2}\right) & C_{\mathrm{m}}\left(\theta_x-\frac{\pi}{2}\right) & 0 & 0 \\ C_{\mathrm{sl}}+C_{\mathrm{i}} & -C_{\mathrm{l}} & -C_{\mathrm{i}} & 0 \\ -C_{\mathrm{l}} & C_{\mathrm{sl}}+C_{\mathrm{i}} & 0 & -C_{\mathrm{i}} \\ -C_{\mathrm{i}} & 0 & C_{\mathrm{i}} & 0 \\ 0 & -C_{\mathrm{i}} & 0 & C_{\mathrm{i}} \end{bmatrix} \tag{1}$$

$$C_{\mathrm{m}}(\theta_x) = -C_{\mathrm{m0}} - C_{\mathrm{m1}}\cos(\theta_x), \tag{2}$$

$$\theta_x = \frac{\pi}{2p}x \tag{3}$$

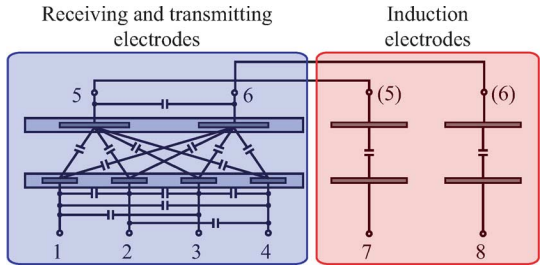


Fig. 2. Equivalent circuit model for analysis [16].

where $C_{\mathrm{st}}, C_{\mathrm{sl}}, C_{\mathrm{ta}}, C_{\mathrm{tb}}, C_{\mathrm{l}}, C_{\mathrm{i}}, C_{\mathrm{m0}}$, and C_{m1} are all positive constants determined by the geometrical relationship among the terminals. The variable x is the slider displacement and θ_x is its electric angle representation where $4p$ of x, which is the repeating cycle of the structure, is equivalent to 2π of θ_x.

Among these coefficients, only $C_{\mathrm{m}}(\theta_x)$ is the function of the slider position and is approximated by sinusoidal function; the other coefficients are simply approximated by constants. These approximations are based on the experimental results of coefficients measurements, which are described in the later section.

Using the capacitance coefficients matrix, the relationship between charges (or currents) and the voltages on the terminals can be expressed as

$$\mathbf{q} = \mathbf{CV} \tag{4}$$

where $\mathbf{q}$ and $\mathbf{V}$ are 8×1 vectors that represent the charges and the voltages on the eight terminals, respectively.

C. Signal Analysis

In this section, we analyze the voltages that are received at the four-phase electrode on the stator. In our encoder setup, excitation voltages are applied to the induction electrodes on the stator, which are terminals 7 and 8. Therefore, the voltage vector $\mathbf{V}$ becomes

$$\mathbf{V} = \{V_1, V_2, V_3, V_4, V_5, V_6, V\sin\omega t, -V\sin\omega t\} \tag{5}$$

where V and ω are the voltage amplitude and the angular frequency of the applied voltages to the induction electrodes. Voltages V_1 through V_6 are unknown.

Next, we define the charge vector $\mathbf{q}$ as

$$\mathbf{q} = \{0, 0, 0, 0, 0, 0, q_7, q_8\} \tag{6}$$

where q_7 and q_8 are the charges given to the induction electrodes by the signal source. To determine the charges on the terminals 1 through 4, we introduced an assumption; the voltages of the four-phase electrode will be measured using high input impedance circuit and thus no electric charge will flow into or out from the electrode. Due to this assumption, the charges on the terminals 1 through 4 become zero. The charges on the terminals 5 and 6 are also determined as zero, since they are not connecting to any electric power source.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

Substituting (1), (5) and (6) for (4) derives the induced voltages V_1, V_2, V_3 and V_4 as

$$\begin{cases} V_1 = K_c V \cos(\theta_x)\sin(\omega t) \\ V_2 = K_c V \sin(\theta_x)\sin(\omega t) \\ V_3 = -K_c V \cos(\theta_x)\sin(\omega t) \\ V_4 = -K_c V \sin(\theta_x)\sin(\omega t) \end{cases} \tag{7}$$

where K_c is

$$K_c = \frac{2C_{\mathrm{i}}C_{\mathrm{m1}}}{-4{C_{\mathrm{m1}}}^2 + (C_{\mathrm{i}} + C_{\mathrm{l}} + C_{\mathrm{sl}})(C_{\mathrm{st}} + C_{\mathrm{tb}})}. \tag{8}$$

Signals in (7) are amplified as

$$V_{\mathrm{A}} = K_o(V_1 - V_3) = 2K_oK_c\cos(\theta_x)\sin(\omega t) \tag{9}$$

$$V_{\mathrm{B}} = K_o(V_2 - V_4) = 2K_oK_c\sin(\theta_x)\sin(\omega t) \tag{10}$$

where K_o is the amplification ratio of the differential amplifiers. Differential amplification can remove common mode noise from receiving signals; therefore, the noise tolerance of the encoder can be improved.

The signals in (9) and (10) are amplitude-modulated signals. Demodulating these signals gives

$$V_{\mathrm{A}} = A\cos(\theta_x) \tag{11}$$

$$V_{\mathrm{B}} = A\sin(\theta_x) \tag{12}$$

where A is the eventual amplification constant after (9) and (10) are multiplied and filtered by a synchronous demodulator. Finally, the slider position θ_x is calculated as

$$\theta_x = \arctan\frac{A\sin(\theta_x)}{A\cos(\theta_x)}. \tag{13}$$

III. Optimization of Electrodes

In this section, electrode geometries are optimized in terms of the output error. In the analytical model described in the previous section, the capacitance changes were approximated by sinusoidal waves as in (2). Practically, the capacitance changes are not pure sine waves; they contain harmonics that lead to the error in the encoder output. The changes of capacitances are easily affected by, e.g., the geometry of electrodes or the gap between the slider and the stator [18]. Therefore, in this section, changes of capacitances will be evaluated for different electrode sizes and different gaps between the slider and the stator.

The analyses are performed using the surface charge method. The encoder model used in the analyses is shown in Fig. 3(a). Since the analyses focused on the capacitance changes between the two-phase and the four-phase electrodes, the induction electrodes are omitted in this model. For ease of calculations, substrates of sensor electrodes are ignored. The medium between the slider and the stator was considered as air. The pitch p was normalized to 1. The proportion of electrode width against the pitch p is defined as w. Similarly, the proportion of the gap against the pitch is defined as d.

Although the induction electrodes are omitted in the simulation model, the capacitance coefficients C_{i} of induction electrodes is necessary for the output calculation. Therefore, the coefficients C_{i} were calculated as parallel-plate capacitors that has electrodes' gap d and total electrode width t as shown in Fig. 4.

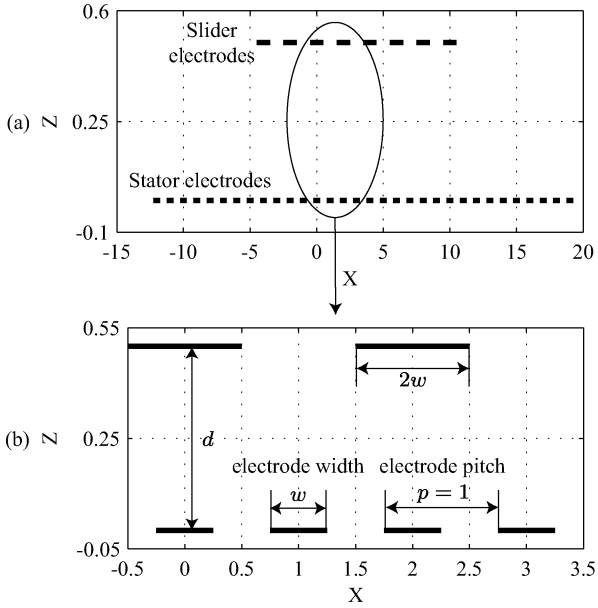


Fig. 3. Encoder model used in analyses of the Section III.

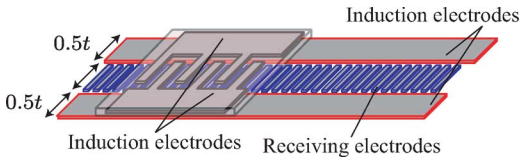


Fig. 4. Definition of the width of induction electrodes.

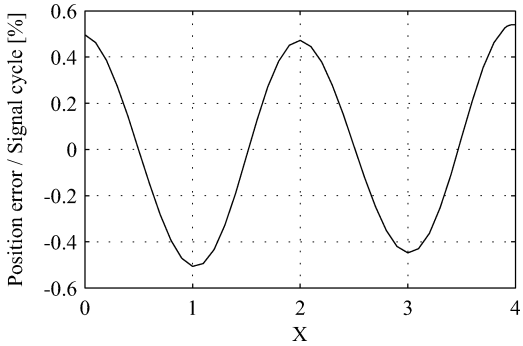


Fig. 5. Simulated position error for one signal cycle when the width w is 0.5.

A. Optimization of the Electrode Width

The electrode width w was changed from 0.1 to 0.9 with an interval of 0.01. The gap between the slider and the stator was fixed to $d = 1.0$. The calculated capacitance coefficients were substituted into (1) and the encoder output was evaluated using (4)–(13). Amplitudes of applied voltages, which were described as V in (5), were set to 1. Fig. 5 shows the position error within one signal cycle when the width w was 0.5. The percentage of the maximum position error against the signal cycle (hereinafter referred to as the periodical error) is about $\pm 0.5\%$ in Fig. 5. Fig. 6 shows periodical errors for different widths. The plot shows that the periodical error is minimized at between 0.4 and 0.5 of w. The graph also shows that the position error does not depend on the area of induction electrodes.

The output was also evaluated from the viewpoint of the signal amplitude, since the amplitude of the raw output signal

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

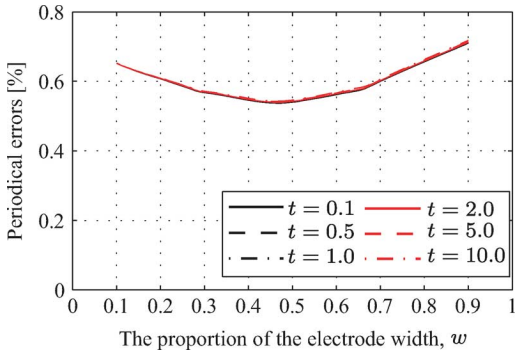


Fig. 6. Simulated periodical errors for different electrode widths.

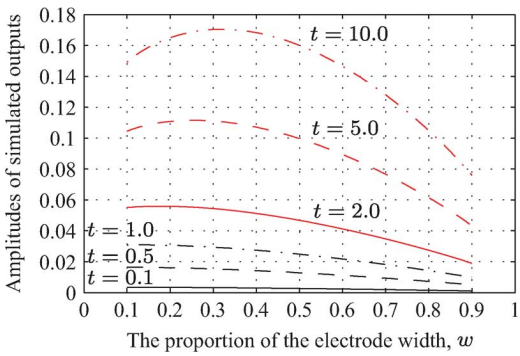


Fig. 7. Simulated amplitudes of sensor outputs for different electrode widths.

directly affects the signal-to-noise ratio. Fig. 7 shows amplitudes of the output signals. Unlike the case of the position error, the amplitude depends on the area of induction electrodes. The plot shows wider induction electrodes result in bigger amplitudes. The plot also shows that amplitudes decrease after peaks around $w = 0.3$. Although, larger amplitude is better for noise tolerance, the change of amplitude is not so drastic within $w = 0.1$ to 0.6. Therefore, it can be concluded that the optimum electrode width is between 0.4 and 0.5 of w that minimizes the periodical error.

B. Optimization of the Gap

Next, we evaluated the effect of the gap between the slider and the stator. The proportion of the gap, d, is changed from 0.1 to 3.5 with an interval of 0.1. For this analysis, the proportion of electrode width, w, was fixed to 0.5. Then, calculated capacitances were also substituted into (1) and the encoder output was evaluated using (4) through (13). Calculated periodical errors are shown in Fig. 8. The plot shows the periodical error is minimized with the gap d is between 0.5 and 1.5. In addition, it can be found that the error does not depend on the area of induction electrodes.

Fig. 9 shows the change of amplitudes in simulation outputs. The amplitude depends on the area of induction electrodes. The graph also shows that bigger gap results in lower amplitudes. Although smaller gap is better from the viewpoint of signal-to-noise ratio, the position error gets worse as the gap becomes smaller than 0.5. These results indicate the desired range of the gap is between 0.5 and 1.5.

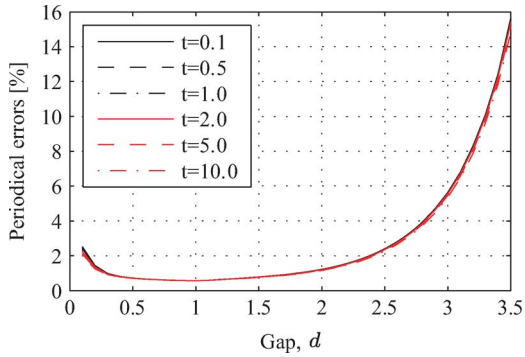


Fig. 8. Periodical errors of the simulation results for different gaps.

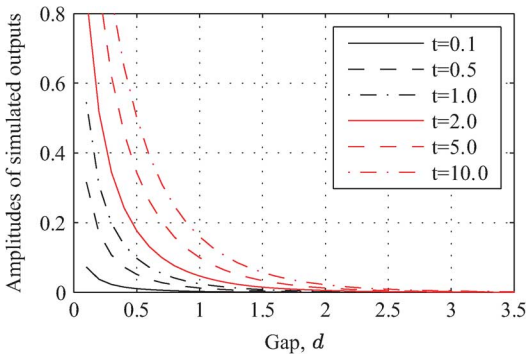


Fig. 9. Simulated amplitudes of sensor outputs for different gaps.

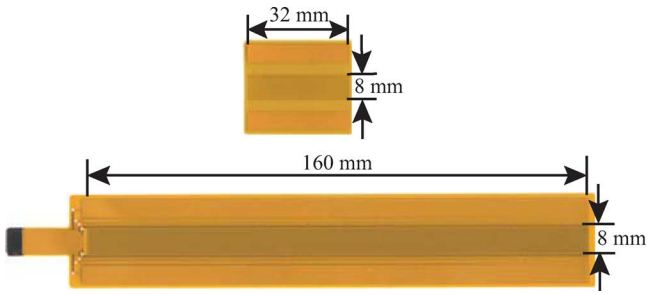


Fig. 10. Photograph of slider (upper) and stator (lower) films.

IV. Fabrication of a Prototype

A. Electrode Films

The slider and stator films of the prototype system are shown in Fig. 10. Both of them are fabricated by flexible printed circuit (FPC) technology. Their thickness measures 0.2 mm, which allows the sensor to be setup in thin interspaces or on curved surfaces. The pitch of the receiving electrodes is 200 μm and that of the transmitting electrodes is 400 μm. The proportion of the electrode width against the pitch is set to 0.5. The total area of the receiving electrodes on stator measures 8 mm wide by 160 mm long, whereas the area of the transmitting electrodes on the slider measures 8 mm by 32 mm. All the induction electrodes have a width of 6 mm and their lengths are 162 mm for the stator and 32 mm for the slider. All electric terminals are integrated in the stator; hence, no terminals are placed on the slider.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

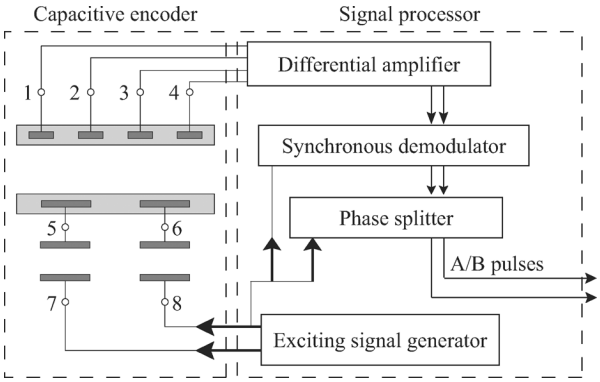


Fig. 11. Electronic interface for the capacitive encoder.

B. Electric Circuit and Signal Processing

Fig. 11 shows the diagram of the electric interface developed for this prototype encoder. The interface consists of differential amplifiers, synchronous demodulators, a phase splitter and a signal generator.

First, the exciting signal generator generates mirrored sine voltages for induction electrodes, one of which is also used as reference signals for the synchronous demodulator and the phase splitter. The four-phase receiving electrode detects the modulated sine waves corresponding to (7), which are measured and amplified by the differential amplifiers. Then, those signals are demodulated by the synchronous demodulators. Finally, the phase splitter calculates the phase θ_x from the signals. In this particular prototype, the output was converted into A/B pulses.

The phase splitter employed in the system generates 1024 pulses per repeating cycle of the signal. Since the signal θ_x repeats every 800 μm, it gives the pulse resolution of 0.8 μm.

V. Experiments

A. Capacitances

The capacitance coefficients of the prototype were measured using an impedance analyzer as described in [17]. Fig. 12 shows a part of the measurement results of the capacitance coefficients. The other results are not shown in the plot due to the limited space. Plots in Fig. 12(a) and (b) shows the coefficients that were approximated by the sine function in (1). They correspond to capacitances between transmitting electrodes and receiving electrodes. As can be seen, capacitance changes between transmitting electrodes and receiving electrodes show four-phase sinusoidal waves and their phase are mutually inverted [e.g., C_{15} in Fig. 12(a) and C_{16} in Fig. 12(b)], although some offsets were observed among coefficients. The plot in Fig. 12(c) shows the capacitance coefficients that were approximated by constants or zero in (1). They also show some offsets.

From the measurement results, the constants in (8) were identified by averaging corresponding data to find that C_{m1} is 1.58 pF,C_{st} is 174 pF, C_{sl} is 24.6 pF, C_{tb} is 47.5 pF, C_{l} is 29.1 pF, and C_{i} is 35.9 pF.

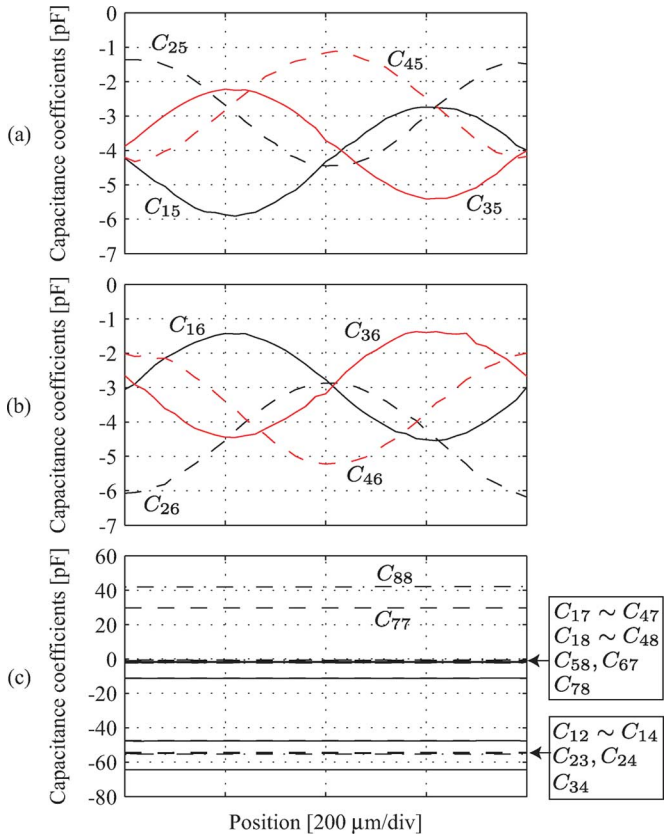


Fig. 12. Measurement results of capacitance coefficients.

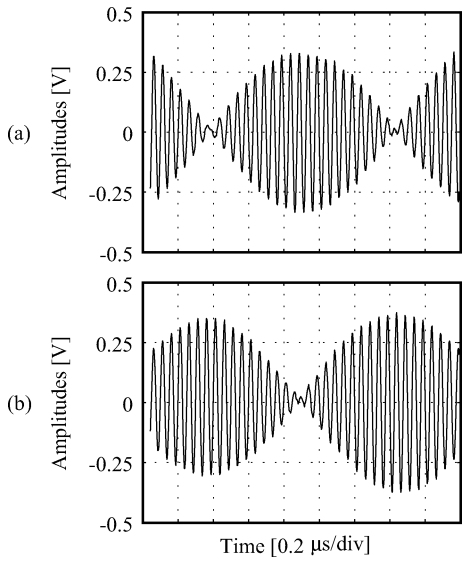


Fig. 13. Output signals of differential amplifiers. Plot (a) is V_{A} and plot (b) is V_{B}.

B. Signals

The measured voltages that correspond to (11) and (12) are plotted in Fig. 13. The amplitude and frequency of the applied excitation voltage are 3.6 $\mathrm{V_{p-p}}$ and 20 kHz. Although amplitudes of output signals should be about 1.0 $\mathrm{V_{p-p}}$ by theoretical estimation, the experimental results are smaller than theoretical

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

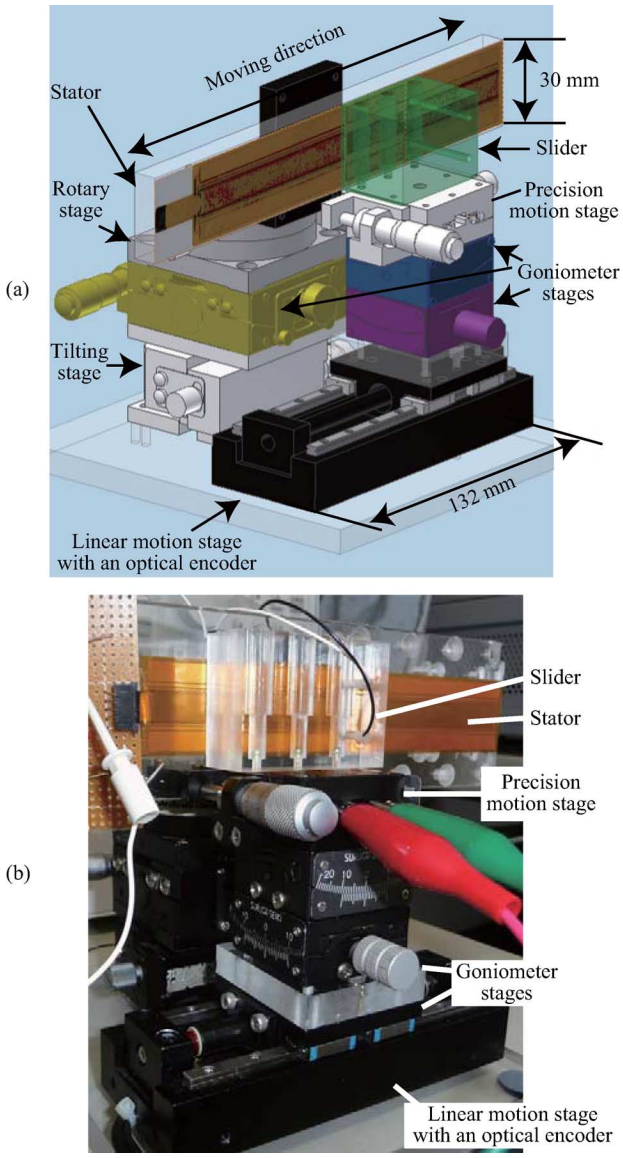


Fig. 14. (a) Schematic diagram of the experimental setup composed of the capacitive encoder and the linear motion stage with an optical linear encoder as a reference. (b) Photograph of the experimental setup.

values. However, it can be clearly seen that the applied voltages are modulated as described in (11) and (12).

C. Linearity

Fig. 14 shows the experimental setup used to test the static characteristics of the prototype encoder. The setup consists of a rotary stage, a tilting stage, goniometer stages, a precision motion stage and a linear motion stage with an optical linear encoder. The rotary, tilting, goniometer and precision motion stages were used to correctly align the slider with the stator. The gap between the slider film and the stator film was kept at about 100 μm. Since the films have cover layers of 12.5 μm, resultant electrode gap is about 125 μm. The linear motion stage was used to give displacement to the slider automatically.

The motion of the slider was controlled by a servo system using the optical linear encoder (MicroE Mercury 3500). In Fig. 14, it is set just under the slider head; therefore, the encoder

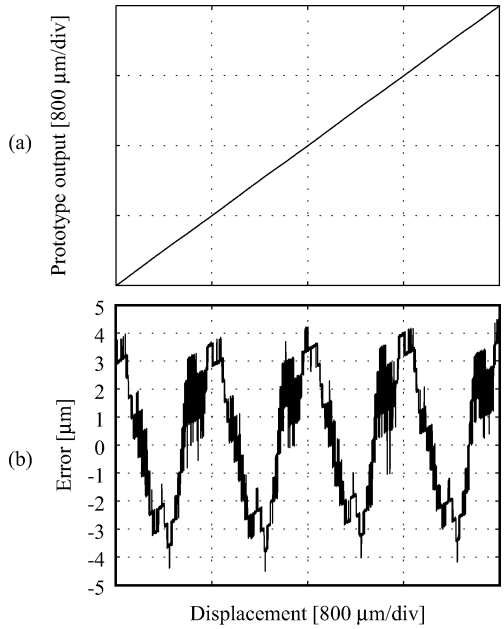


Fig. 15. (a) Output of the capacitive encoder against the optical linear encoder. (b) Linearity error based on the optical linear encoder as a reference.

cannot be seen. The output of the optical encoder was used as a reference value. The measurement range of the optical encoder is 50 mm, the resolution is 0.1 μm, and the accuracy is ± 1 μm over the full scale.

Fig. 15(a) shows the output of the prototype encoder, which was plotted against the output of the optical linear encoder. For this measurement, the slider was moved with a step of 5 μm. This plot shows good linearity of the prototype encoder. Fig. 15(b) shows the error between the output of the optical encoder and the output of the prototype encoder. From this plot, the periodical error that has a repeating cycle of 800 μm can be found. According to this plot, the maximum error of ± 4 μm can be found. Although the pattern of the errors are different between the simulation (Fig. 5) and the experiment, the amplitude of the error (± 4 μm $=$ $\pm 0.5\%$) was found almost the same.

VI. Conclusion

This paper analyzed and tested a new electrostatic capacitive encoder using electrostatic induction. The encoder has a two-phase transmitting and four-phase receiving electrodes and power feeding to the transmitting electrodes is realized by electrostatic induction, thus no wiring is needed to the sliding part. The operation principle was discussed using the capacitance network model. Based on the discussion, the proportion of the electrode width and the gap between the slider and the stator were optimized by numerical simulations. The appropriate width was found to lie between 0.4 and 0.5 of the electrode pitch. In the gap optimization, it can be found that the position error fluctuates little when the gap is changed from 0.5 to 1.5 of the pitch. Finally, the operation was experimentally verified using a fabricated prototype. The experimental results showed the maximum error of ± 4 μm, which is in the same order as the simulation predicted.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] I. Alejandre and M. Artes, "Method for the evaluation of optical encoders performance under vibration," *Precision Eng.,* vol. 31, no. 2, pp. 114–121, Apr. 2007. [Online]. Available: http://dx.doi.org/10.1016/j.precisioneng.2006.03.004

[2] I. Alejandre and M. Artes, "English thermal non-linear behaviour in optical linear encoders," *Int. J. Mach. Tools Manufact.,* vol. 46, no. 12-13, pp. 1319–1325, 2006. [Online]. Available: http://dx.doi.org/10.1016/j.ijmachtools.2005.10.010

[3] Z. Du, C. Lv, and M. Hong, "English research on error modelling and identification of 3 axis nc machine tools based on cross grid encoder measurement," in *Proc. J. Phys. Conf. Series*, 2006, vol. 48, no. 1, pp. 91–100. [Online]. Available: http://dx.doi.org/10.1088/1742-6596/48/1/017

[4] C.-L. Guan, Y.-F. Dai, X.-H. Xie, and Z.-Q. Yin, "English investigation on feedback control of linear motors in ultraprecision-machine feed-drive systems," *J. Vacuum Sci. Technol. B,* vol. 27, no. 3, pp. 1351–1354, May 2009. [Online]. Available: http://dx.doi.org/10.1116/1.3137966

[5] R. K. Heilmann, C. G. Chen, P. T. Konkola, and M. L. Schattenburg, "Dimensional metrology for nanometre-scale science and engineering: Towards sub-nanometre accurate encoders," *Nanotechnol.,* vol. 15, no. 10, pp. 504–511, 2004. [Online]. Available: http://dx.doi.org/10.1088/0957-4484/15/10/002

[6] S. K. Kuo and C. H. Menq, "Modeling and control of a six-axis precision motion control stage," *IEEE/ASME Trans. Mechatron.,* vol. 10, no. 1, pp. 50–59, 2005. [Online]. Available: http://dx.doi.org/10.1109/TMECH.2004.842219

[7] Y. Kakino, Y. Ihara, S. Lin, S. Hayama, K. Kawakami, and M. Hamamura, "Japanese measurement of motion accuracy and improvement of machining accuracy on ultra-high precision nc machine tools by using cross grid encoder test," *Jpn. J. Jpn. Soc. Precision Eng.* vol. 62, no. 11, pp. 1612–1616, Nov. 1996. [Online]. Available: http://ci.nii.ac.jp/naid/110001371993/

[8] L. K. Baxter*, Capacitive Sensors*. New York: IEEE Press, 1997.

[9] *A Simple Capacitive Displacement Sensor*, 0924-4247(91)87003-L, Mar. 1991. [Online]. Available: http://dx.doi.org/10.1016/0924-4247(91)87003-L

[10] W. Heerens, "English multi-terminal capacitor sensors," *J. Phys. E* vol. 15, no. 1, pp. 137–141, Jan. 1982 [Online]. Available: http://dx.doi.org/10.1088/0022-3735/15/1/027

[11] W. C. Heerens, "English application of capacitance techniques in sensor design," *J. Phys. E.* vol. 19, no. 11, pp. 897–906, 1986. [Online]. Available: http://dx.doi.org/10.1088/0022-3735/19/11/002

[12] F. Kimura, M. Gondo, N. Yamashita, A. Yamamoto, and T. Higuchi, "Prototyping of flexible capacitive encoder with untethered slider using electrostatic induction," in *Proc. 7th IEEE Conf. Sensors*, 2008, vol. 1, pp. 1619–1622. [Online]. Available: http://dx.doi.org/10.1109/ICSENS.2008.4716761

[13] G. Masahiko, "Electrostatic encoder and electrostatic displacement measuring method," U.S. Patent 7199727, Apr. 2007.

[14] N. David*, Linear Position Sensors*. New York: Wiley, 2004.

[15] H. Kouji and S. Nobuyuki, "Capacitive-Type Displacement Measuring Apparatus" U.S. Patent 6292001, Sep. 18, 2001. [Online]. Available: http://www.freepatentsonline.com/6292001.html

[16] N. Yamashita, Z. G. Zhang, A. Yamamoto, M. Gondo, and T. Higuchi, "English voltage-induction type electrostatic film motor driven by two- to four-phase ac voltage and electrostatic induction," *Sens. Actuators A,* vol. 140, no. 2, pp. 239–250, 2007. [Online]. Available: http://dx.doi.org/10.1016/j.sna.2007.07.007

[17] A. Yamamoto, T. Niino, and T. Higuchi, "Modeling and identification of an electrostatic motor," *Precis. Eng.* vol. 30, no. 1, pp. 104–113, Jan. 2006 [Online]. Available: http://dx.doi.org/10.1016/j.precisioneng.2005.06.004

[18] P. D. A. T. R. Hicks*, The Nano Positioning Book: Moving and Measuring to Better Than a Nanometre*. London, U.K.: ISTE, 1997.



Fumitaka Kimura (S'09) received the B.S. degree in engineering from Yokohama National University, Kanagawa, Japan, and the M.S. degree in engineering from the University of Tokyo, Tokyo, Japan, where he is currently working towards the Ph.D. degree.

His current research interests include displacement or angle sensors, electrostatic actuators, and mechatronics.



Masahiko Gondo was a Research Developer with Japan Radio Company, Ltd., from 1981 to 1999 and with Olympus Corporation from 1990 to 2005. From 2005 to 2008, he was a Researcher with the University of Tokyo, Tokyo, Japan, and an Entrepreneur of Creation and Support Program for Startups from Universities from Japan Science and Technology (JST), Japan. He was preparing to launch a company in cooperation with the University of Tokyo. Since 2008, he has been a Representative Director with Seidensha Company, Ltd., a venture business which was founded to industrialize his inventions. His current research interests include displacement or angle sensors and mechatronics.



Norio Yamashita (S'07) received the B.S. and M.S. degrees in engineering from the University of Tokyo, Tokyo, Japan.

Since 2008, he has been a Researcher in RIKEN, Saitama, Japan. His research interests include mechatronics, robotics, and precision engineering.



Akio Yamamoto (S'96–A'99–M'03) received the B.S., M.S., and Ph.D. degrees in precision engineering from the University of Tokyo, Tokyo, Japan, in 1994, 1996, and 1999, respectively.

He was a Research Associate with the Department of Precision Engineering, University of Tokyo, from 1999 to 2000, a Lecturer from 2000 to 2005, and since 2005, he has been an Associate Professor. His current research interests include mechatronics, electrostatic actuators, and human–machine interactions.



Toshiro Higuchi (M'87) received the B.S., M.S., and Ph.D. degrees in precision machinery engineering from the University of Tokyo, Tokyo, Japan, in 1972, 1974, and 1977, respectively.

From 1977 to 1978, he was a Lecturer with the Institute of Industrial Science (IIS), University of Tokyo, and an Associate Professor from 1978 to 1991. Since 1991, he has been a Professor in the Department of Precision Engineering, University of Tokyo. He was the Leader of the Higuchi Ultimate Mechatronics Project, Kanagawa Academy of Science and Technology, from 1992 to 1997. His current research interests include mechatronics, magnetic bearings, electrostatic actuators, microelectromechanical systems (MEMS), robotics, and manufacturing.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on November 08,2024 at 13:36:42 UTC from IEEE Xplore. Restrictions apply.